

cutting problems

digitalisierung und programmierung · input, session 1

agenda

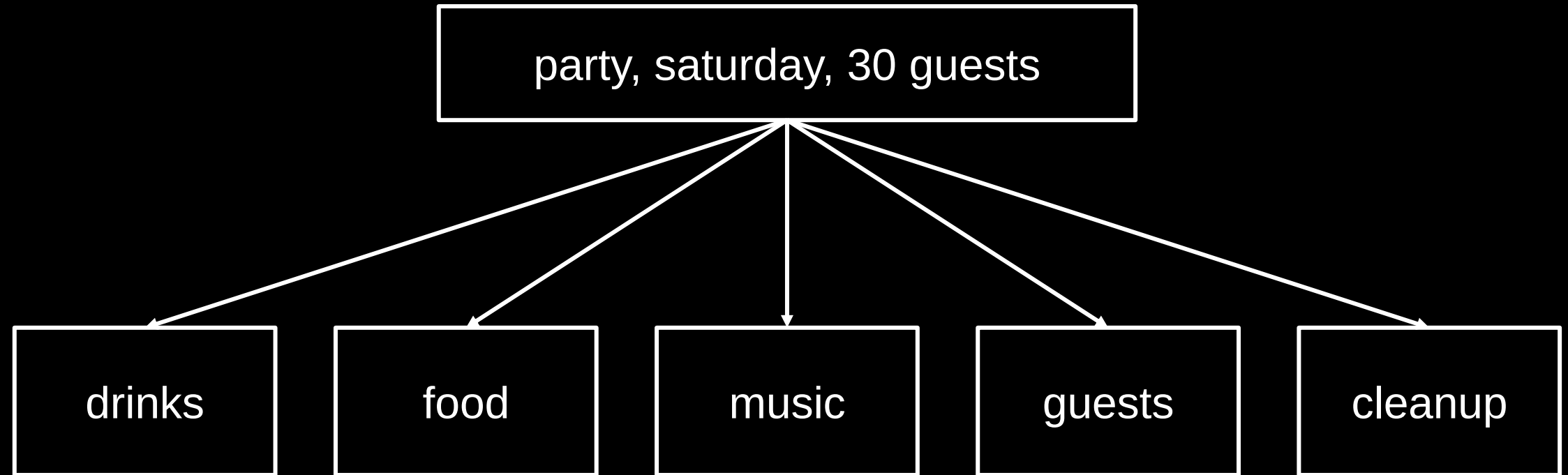
- 1 you can already do this
- 2 this semester's elephant
- 3 three ways to cut
- 4 a good cut is testable
- 5 cuts are agreements

you can already do this

how do you eat
an elephant?



you plan like this without thinking

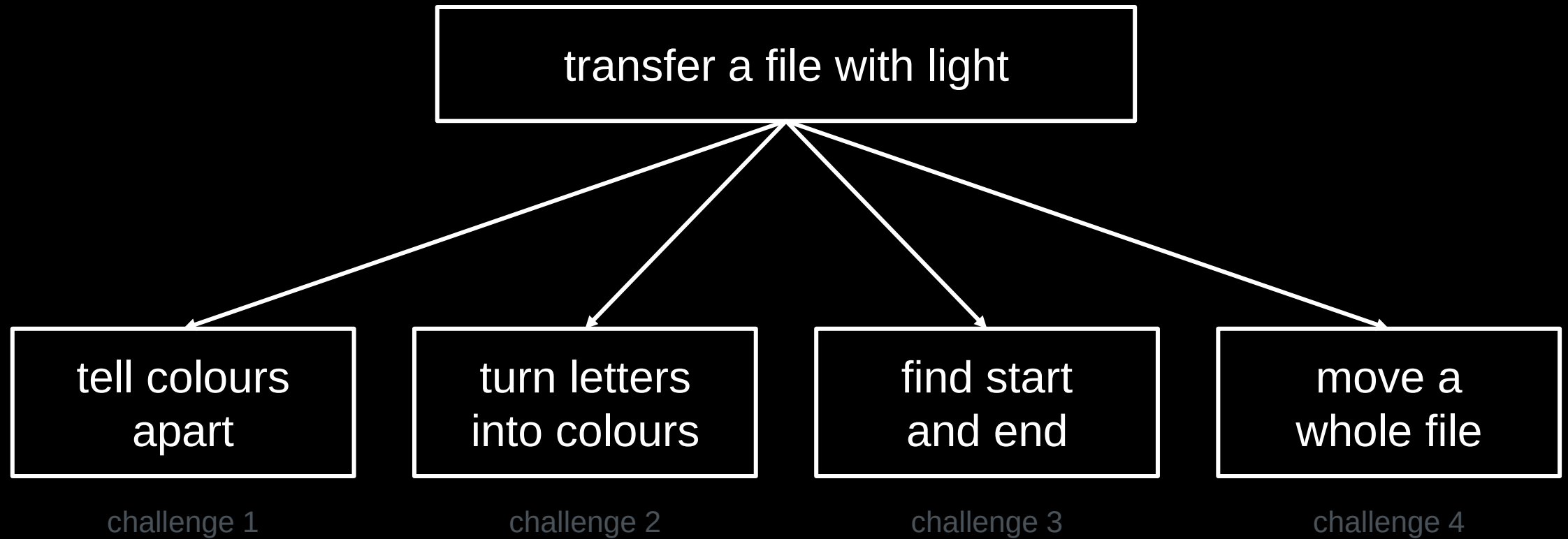


this semester's elephant

this semester's elephant

transfer a file from one laptop
to another, using light.

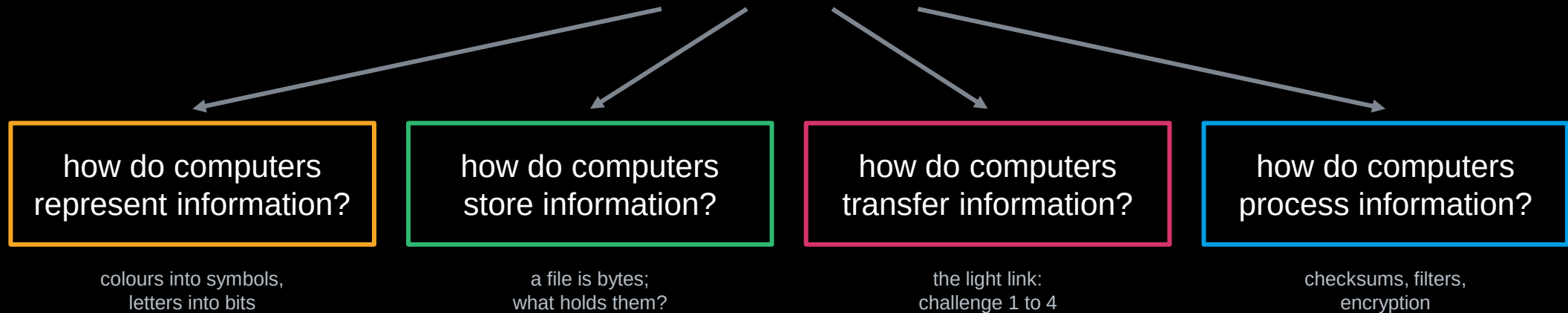
the same problem, cut



one question, four questions

solving problems: cutting them · input, processing, output · measuring · thinking in layers

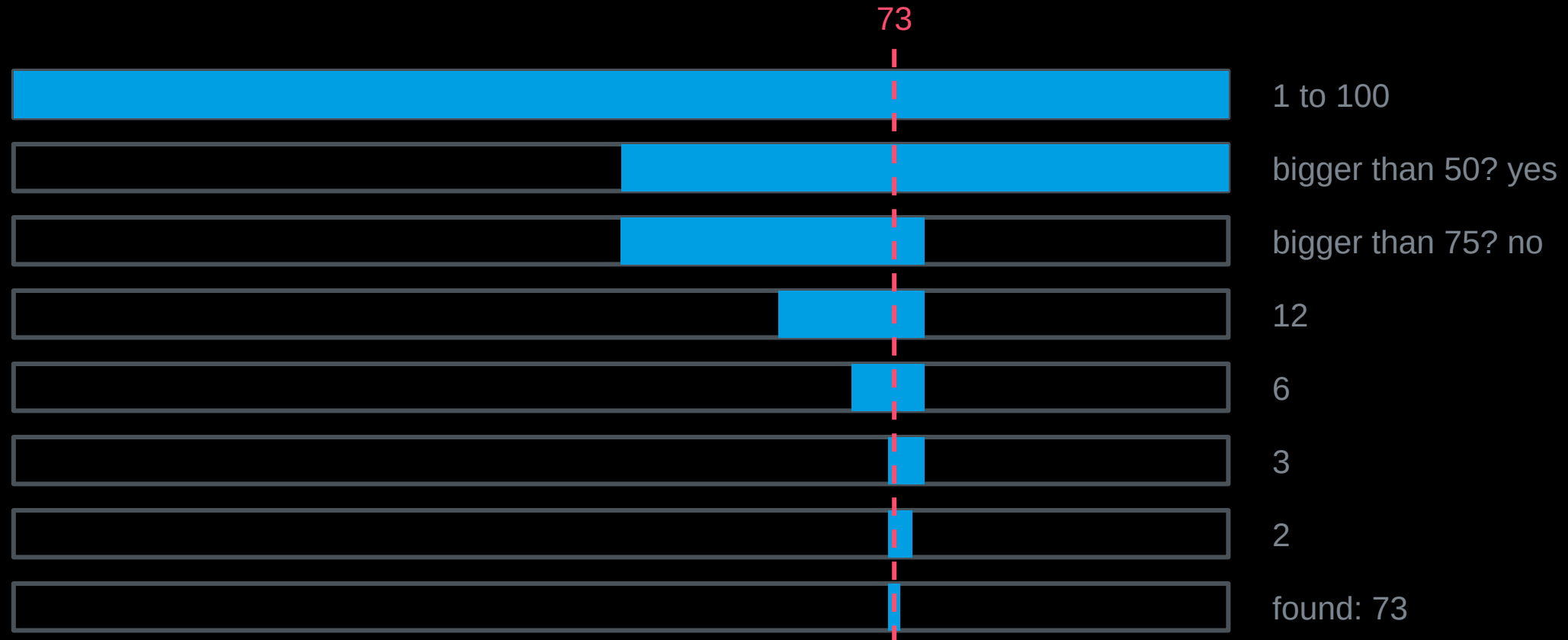
how can we solve complex problems with computers?



one project, four questions. every session works on one of them, and its colour tells you which.

three ways to cut

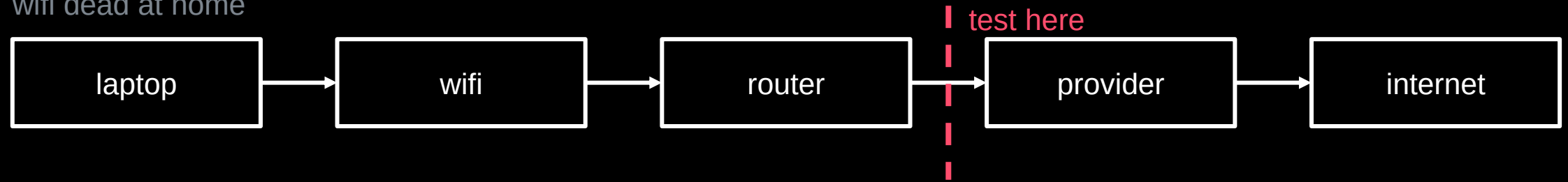
divide & conquer: halve until it's easy



7 questions are enough.
how many steps in general?

the same trick when something breaks

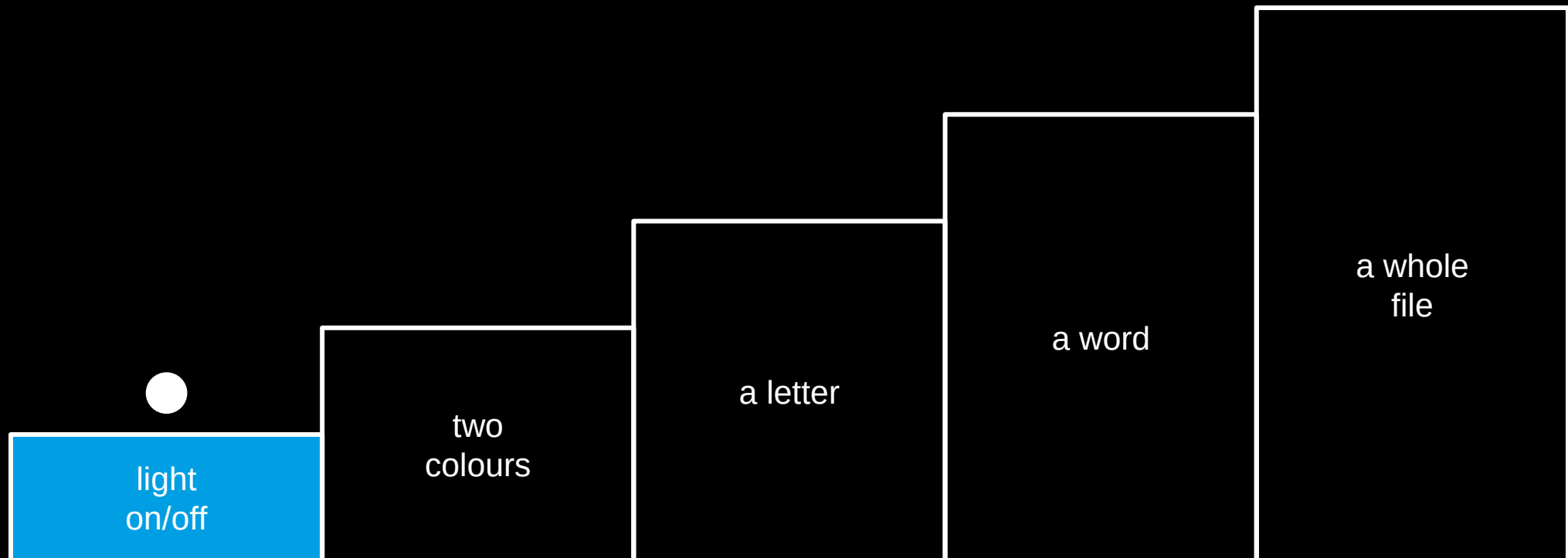
wifi dead at home



nothing arrives in the project



start smaller: solve an easier version first



a good cut is testable

which one are you actually preparing?

i need to learn maths.

i can pass the 2023 exam
in 90 minutes.



the same, in this project

the led works.

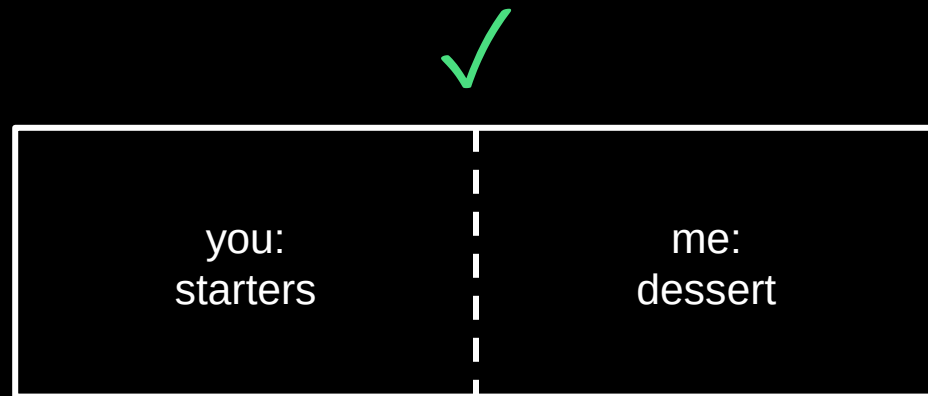
red or blue?
right 50 times in a row.



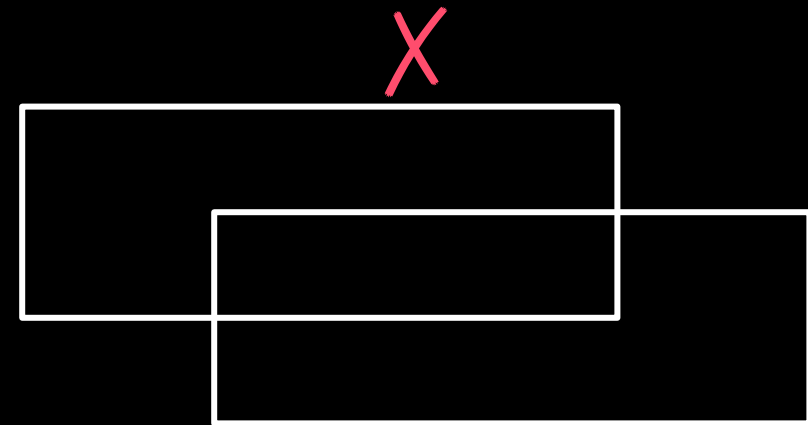
done means: the test passes.

cuts are agreements

where you cut decides how much you must talk



works



we both kind of do the food

and this is how you talk to the machine

one big wish

```
build me the  
file transfer.
```

one small piece, with its test

```
read the sensor ten times,  
return the average.  
i test it: led off, led on.
```

you cut. the machine solves pieces. you check.